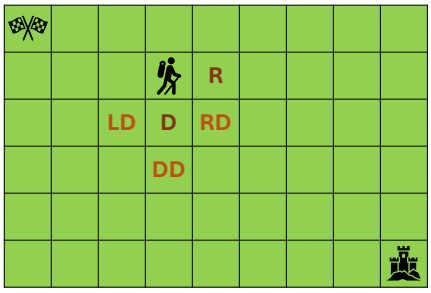
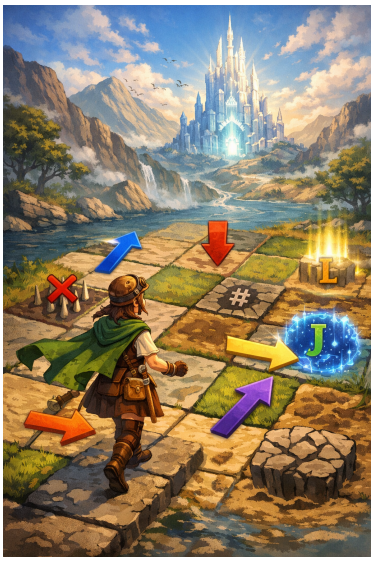
Reaching the Crystal Castle

The legendary explorer Ava seeks to reach the Crystal Castle, located at the far corner of a magical grid-shaped land of tiles. The land is treacherous: some tiles are blocked by quicksand or have magical barriers that restrict how she can move from them.

Ava can move in five ways: she can make a normal step to the right (R) or downward (D), and she can leap diagonally left-down (LD), just downwards (DD) or diagonally right-down (RD), as it is shown in the figure below.





Designed by ChatGPT

However, Ava knows her limits. Jumping (diagonally or just downwards) tires her, so she cannot jump too many times consecutively, nor can the number of jumps in the entire journey exceed a certain amount. Moreover, the magical land has special tiles: some tiles forbid diagonal leaps or all kinds of jumps, because of magical barriers, and some tiles cannot be stepped on, due to quicksand.

Task

Write a program that, given a description of the magical land and Ava's limits, computes the number of paths that Ava can take from the top-left tile to the bottom-right tile (where the Crystal Castle is located). Since the number of paths may be large, output it modulo $10^9 + 7$.

It is guaranteed that, for the given inputs, the top-left and the bottom-right tiles can be stepped on.

Input

On the input first line there is an integer, T, which represents the number of test cases.

For each test case there is a first line containing four integers, R, C, M, and N, which denote the number of rows and the number of columns of the magical land, the maximum number of consecutive jumps and the maximum number of jumps that Ava can make in the entire journey, respectively. Then, R lines follow, each with C characters. Each character indicates the characteristics of the corresponding tile:

- `.` (A tile with no constraints.)
- `X` (A tile from which diagonal jumps (left-down or right-down) are forbidden.)
- `J` (A tile from which jumps (left-down, just downwards or right-down) are forbidden.)
- `#` (A tile that cannot be stepped on.)

Constraints

$1 \leq T \leq 20$	Number of test cases
$1 \leq R \leq 400$	Number of rows
$1 \leq C \leq 400$	Number of columns
$1 \leq M \leq 5$	Maximum number of consecutive jumps
$1 \leq N \leq 10$	Maximum number of jumps in a journey

Output

The output consists of T lines, each with a single integer, p_i, which represents the number of paths Ava can take, in the ith test case, from the top-left tile to the bottom-right tile, modulo $10^9 + 7$ (for every $i = 1, \ldots, T$).

Sample Input

```
3
3 4 1 3
.X.J
..#.
#...
3 4 2 3
.X.J
..#.
#...
10 20 5 10
....................
....................
....................
....................
....................
....................
....................
....................
....................
....................
```

Sample Output

```
12
15
140916123
```

Sample Explanation

In the first test case, Ava cannot make more than one consecutive jump. The 12 possible paths, defined by the sequences of step types, are the following:

D	R	D	R	R	
D	R	RD	R		
D	RD	R	R		
R	D	D	R	R	
R	D	RD	R		
R	DD	R	R		
R	R	DD	R		
R	R	LD	D	R	R
R	R	R	D	D	
R	R	R	D	LD	R
R	R	RD	D		
RD	D	R	R		

In the second test case, Ava can make two consecutive jumps. Besides the possible paths of the first test case, there are three more alternatives:

R	R	LD	RD	R
R	R	RD	LD	R
RD	RD	R		